

Course Syllabus

Biol 413 lab Summer 2025

Contact Information:

Keith Legro

Email : keith.legro@unh.edu

Lab Schedule:

Pre Module	Scientific Investigation Power Point	May 19-22
Module 1	Stickleback Evolution Virtual Lab	May 23-29
Module 2	mitosis/meiosis	May 30-Jun 3
Module 3	Blood Typing	Jun 4-9
Module 4	Immunology (ELISA)	June 10-13
Module 5	CSI Wildlife	Jun 14-17

Grades:

Worksheet on Scientific Method	25pts
Blood typing worksheet	40pts
Worksheet on Elephant poaching CSI	30pts
Quizzes	30 pts

Structure:

In this lab you will complete 5 modules and a Pre Module all of which consist of different interactive labs or information. What to do for each one is listed in brief detail below.

Pre Module

This module is information on the Scientific Method. It deals with the over arching theme that will run through all the modules you will complete in this lab. It is important that you read and understand it before completing the modules. There will be a worksheet to fill out here on Canvas on this material.

Module 1 Stickleback Evolution Lab

For this module you will complete the Stickleback Evolution virtual lab. You will also want to read over the file on Chi square testing and complete the practice Chi square tests in the module as those will help you better complete the simulation. Once the virtual lab is completed you will have a quiz to take on the material.

Module 2 mitosis/meiosis

For this module there is an interactive where you will work through Mitosis as well as a video on mitosis and meiosis with beads to watch. There is also a file with an outline of cell division and the cell cycle that you also will want to read. There is no worksheet for this module however there will be a timed quiz here on Canvas on the material covered.

Module 3 Blood Typing

In this module you will familiarize yourself with blood types and the genetics behind them.

Module 4 Immunology (ELISA)

For this module you will want to work through the interactive on ELISA as well as read the document on ELISA. There will be a quiz on this material.

Module 5 CSI Wildlife

For this module you will work through the interactive on Elephant poaching CSI and then fill out and turn in the worksheet that goes with the interactive. You only need to complete case number one and also only need to fill in the worksheet for case one as well. You are however welcome to complete case two also if you wish but it is not required.

Course Summary:

Date	Details	Due
Fri May 23, 2025		Assignment Hypothesis development worksheet (turn in) due by 11:59pm
Fri May 30, 2025		Quiz Stickleback Evolution Quiz
Wed Jun 4, 2025		Quiz Mitosis Quiz
Tue Jun 10, 2025		Assignment Blood Typing assignment
Fri Jun 13, 2025		Quiz ELISA 2020 Quiz
Wed Jun 18, 2025		Assignment CSI Wildlife worksheet